MIL-M-43757A
30 July 1984
SUPERSEDING
MIL-M-43757
6 October 1971

MILITARY SPECIFICATION

MODIFICATION KIT, PACKBOARD, RADIO CARRYING

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the design, materials and fabrication of a modification kit consisting of two cargo support metal shelves and four nylon webbing cargo securing straps for use with metal packboard.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-T-285	-	Thread, Polyester
DDD-L-20	-	Label: For Clothing, Equipage and Tentage (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8465

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MILITARY

MIL-E-480 - Enamel, Baking, Phenol - Or Urea-Formaldehyde
MIL-C-13924 - Coating, Oxode, Black, For Ferrous Metals
MIL-W-43668 - Webbing, Textile, Bulked Nylon
MIL-P-43747 - Plate, Packboard; and Shelves, Cargo Support, Rucksack Frame

STANDARDS

FEDERAL

FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

DRAWING

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

2-3-229 - Strap, Packboard - Cargo, Securing

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 633 - Electrodeposited Coatings of Zinc on Iron and Steel
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this document may appear in the sample, in which case this document shall govern.

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2 and 6.4).

3.3 Components. The modification kit shall consist of the following listed items:

Nomenclature	Quantity required
Shelf, Packboard, Upper	1 each
Shelf, Packboard, Lower	1 each
Strap, Packboard	4 each

3.4 Materials. Materials when specified shall conform to the applicable documents and requirements specified herein.

3.4.1 Webbing, nylon. The 1-inch nylon webbing shall conform to type III, Olive Drab 7 of MIL-W-43668.

3.4.2 Thread, polyester. The thread shall be type I, class 1, sub-class A or B, size F of V-T-285.

3.4.2.1 Color. The thread shall be dyed Olive Drab S-1 (C.A. 66022), and shall show fastness to weathering equal to or better than the standard sample. If no standard sample is available, the thread shall show good fastness to weathering.

3.4.3 Fastener, steel. The steel fastener shall conform to Arizona Gear and Manufacturing Co., Part No. K415, Kev Lok; Acorn Metal Products Co., Part No.'s C-103 and A-101; G.E. Prentice Manufacturing Co., Part No. 3837, State Recreation Inc. Part No. SR-1, or equal.

3.4.3.1 Finish. The rivets shall be given a black oxide coating conforming to class 2, 3 or 4, as applicable, of MIL-C-13924. The two lever components of the fastener shall be given a phosphate treated zinc plate conforming to type IV finish SC3, of ASTM B 633 followed by baked enameling. The baked enamel shall conform to type I or II of MIL-E-480, color black. The coating on all components shall be performed prior to assembly and shall level out to a smooth and uniform film without orange peel, wrinkles, drops, streaks or areas of no film. (See 4.4.1.1).

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3.4.4 Shelf, packboard, upper and shelf, packboard, lower. The upper and lower packboard shelves shall conform to items 2 and 3 of MIL-P-43747.

3.5 Construction. The construction of the modification kit strap components shall conform in all respects to Drawing 2-3-229 and as specified herein.

3.5.1 Stitching, machine. All stitching except bartacking shall conform to FED-STD-751, type 301 with 6 to 8 stitches per inch.

3.5.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1 inch. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the material sewn. The lock shall be imbedded in the materials sewn.

3.5.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

b. Thread breaks, or two or more consecutive skipped or run-off stitches noted during inspection of the packboard strap shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs the ends of the stitching are not required to be backstitched.

3.5.1.2 Automatic stitching. Automatic stitching may be used to perform the required stitch patternms provided the requirements for the stitch pattern, stitches per inch, size and type of thread are met; and at least three or more tying, overlapping or back stitches are used to secure the ends of the stitching.

3.5.1.3 Thread ends. All thread ends shall be trimmed to a length of not more than 1/4 inch.

3.5.2 Fusing of nylon webbing ends. All ends of nylon webbing shall be fused. The apparatus used to fuse the webbing ends shall be capable of providing sufficient heat to provide a smooth edge with the cut ends of the webbing yarns all fused together.

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3.6 Replacement of defective components. During the spreading, cutting and manufacturing process, strap components having material defects or damages that are classified as defects in 4.4.3 shall be removed from production and replaced with non-defective and properly matched components.

3.7 Marking. The identification marking on the strap shall be applied in the location shown on the drawing and shall conform to type IV, class 5 of DDD-L-20. The letters "US" shall be applied in the location and in the size characters indicated on the drawing and shall conform to type IV, class 9 of DDD-L-20. Fastness of the class 9 marking shall be as specified for class 5 marking.

3.8 Workmanship. The components of the modification kit shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4. The presence of any defect shll be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Certification. The contractor shall furnish a certificate of compliance for the finish specified in 3.4.3.1 for the steel fastener.

4.4.2 Shelves, upper and lower. The upper and lower packboard shelves shall be examined for visual and dimensional characteristics in accordance with the quality assurance provisions of MIL-P-43747.

4.4.3 Packboard strap visual examination. The packboard strap shall be examined for the defects listed in table I. The lot size shall be expressed in units of one packboard strap. The sample unit shall be one packboard strap. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0.

TABLE I. Packboard strap visual defects

Examine	Defect
Webbing	Frayed or scalloped edges, not firmly and tightly woven. Ends not fused or not fused as specified, i.e., yarn ends not fused together or not smooth. Free end not cut on a bias.
Hardware	Broken or malformed; corroded areas, burrs or sharp edges. Finish omitted or not as specified. Major components of fastener not free to turn.
Stitches and stitchings	Any row of stitching missing. Number of stitches per inch less than minimum or more than maximum specified. Thread breaks, or two or more consecutive skipped stitches or run-offs not overstitched. Ends of stitching not secured as specified. Tension loose resulting in loose top or bottom thread or tight tension causing puckering of material. Stitch type not as specified.
Construction and assembly	Fastener not attached to webbing as specified. Needle chews.

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TABLE I. Packboard strap visual defects (cont'd)

Examine	Defects
Marking	Omitted, incorrect, illegible, or size of characters not as specified.
Cleanness	Grease or oil stains clearly noticeable. Thread ends not trimmed as specified.

4.4.4 Packboard straps dimensional examination. The packboard straps shall be examined for conformance to the dimensions specified on the drawing. Any dimension not within the specified tolerance shall be classified as a defect. The sample unit shall be one packboard strap. The lot size shall be expressed in units of packboard straps. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

4.4.5 Packaging inspection. An examination shall be made to determine that preservation, packing and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Examination of closure defects listed below shall be made on shipping containers fully packaged. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing. Any component not as specified. Any component damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, improper taping, loose strapping, or inadequate stapling. Bulged or distorted container.
Content	Number of unit packs per shipping container is more or less than required. Number of required components per unit pack is more or less than required. 1/

1/ For this defect, the unit packs from each shipping container in the sample shall be examined.

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4.4.6 Palletization inspection. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limit.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. The modification kit, consisting of four straps and two cargo shelves, shall be compactly arranged and then unit packed in a fiberboard box. Prior to unit packing, the webbing of each strap shall be firmly wound around the hardware component and restrained from unwinding by securely tying with cotton tape or twine. The fiberboard box shall conform to style optional, type CF (variety SW), class domestic, grade 125 of PPP-B-636. Inside dimensions of each box shall approximate 9 inches in length, 3 inches in width and 2-1/4 inches in depth. Approximate dimensions are furnished as a guide only. Box closure shall be secured with 2 inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45.

5.1.2 Commercial. The modification kit shall be preserved in accordance with ASTM D 3951.

5.12 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing. Forty-eight modification kits, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A unit packs shall be packed on end eight in length, six in width and one in depth within a shipping container.

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Inside dimensions of each shipping container shall approximate 22 inches in length, 21 inches in width and 9-1/2 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.2 Level B packing. Forty-eight modification kits, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A unit packs shall be packed on end eight in length, six in width and one in depth within a shipping container. Inside dimensions of each shipping container shall approximate 22 inches in length, 21 inches in width and 9-1/2 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.2.1 Weather-resistant fiberboard. When specified (see 6.2), the shipping containers shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.3 Commercial packing. The modification kit, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), modification kits, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the bonding means K and L or film bonding means O and P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The modification kit, which consists of two cargo support metal shelves and four nylon webbing cargo securing straps with metal fasteners, is intended for use with a metal packboard which in turn is used by personnel for carrying the AN/PRC-77 radio set with TSEC-KY-38.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. When a first article sample is required (see 3.2, 4.3 and 6.4).
c. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
d. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
e. When palletization is required (see 5.3).

6.3 Samples. For access to samples, address the contracting officer issuing the invitation for bids.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The first article should consist of one complete set of the components listed in 3.3. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.5 Source of supply. For the fastener specified in 3.4.3 is AGM Cargo-Ties, Inc., 3544 Each Fort Lowell Road, P.O. Box 6089, Tuscon, AZ,. 85716.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous isue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - NU
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Army - MD
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL–M–43757A	Modification Kit, Packboard, Radio Carrying

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP, 1 Apr 84

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